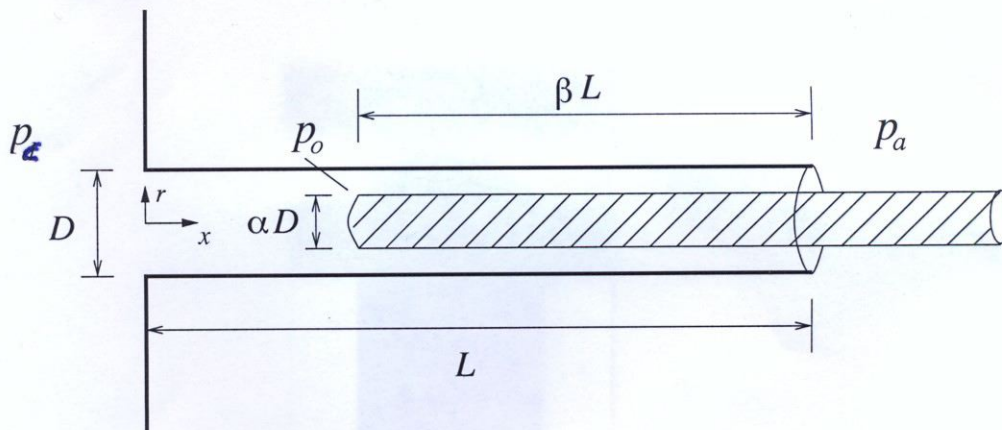


VF4

The liquid ~~contained in a deposit~~ ^{LOCATED IN A CONTAINER} at pressure p_c discharges to the atmosphere, where the pressure is $p_a < p_c$ through a cylindrical pipe of diameter D and length $L \gg D$. To control the volume flux Q that circulates, a cylindrical bar of diameter αD is inserted coaxially from the exit, as shown in the figure, to partially block the liquid flow in the final stretch of the pipe.

1. Give the criterion for the motion in the tube to be dominated by viscosity.
2. Determine, in terms of the value of the pressure at the end of the tube p_o , the velocity field found for $0 < x/L < (1 - \beta)$ and for $(1 - \beta) < x/L < L$, where βL is the insertion length.
3. Obtain the volume flux Q and the value of p_o as a function of β .



- 1) THE MOTION IS STEADY. CONVECTION IS NEGLECTIBLE IF $\frac{\rho V_L D}{\mu} \ll 1$. TO ESTIMATE V_L WE ASSUME THAT VISCOSITY IN FACT DOMINATES, SO THAT $\frac{\Delta p}{L} \sim \mu \frac{V_T}{D^2} \rightarrow V_T \sim \frac{D^2}{L} \frac{\Delta p}{\mu} \sim \frac{D^2}{L} \frac{p_c - p_a}{\mu}$

IN TERMS OF KNOWN QUANTITIES, THE CRITERION TO BE SATISFIED IS

$$\frac{\rho D^4}{\mu^2 L^2} (p_c - p_a) \ll 1$$

2) IF THE ABOVE CONDITION HOLDS, THEN WE FIND DEVELOPED FLOW EVERYWHERE ALONG THE PIPE, EXCEPT IN SMALL REGIONS NEAR THE ENTRANCE AND ALSO NEAR ^{THE TRANSITION REGION} $x = L(1 - \beta)$, BUT THE ASSOCIATED PRESSURE VARIATIONS ARE SMALL, SO WE CAN NEGLECT THEM IN THE FIRST APPROXIMATION AND USE $p(0) = p_c$ TOGETHER WITH A VALUE $p = p_o$ AT THE END OF THE FIRST STRETCH EQUAL TO THAT FOUND AT THE ENTRANCE OF THE SECOND STRETCH.

$$0 < x < (1 - \beta)L \rightarrow v_x = \frac{1}{4\mu} \frac{dp}{dx} \left[r^2 - \left(\frac{D}{2} \right)^2 \right] \Rightarrow Q = \frac{p_c - p_o}{(1 - \beta)L} \frac{\pi D^4}{128\mu} \quad (1)$$

$0 = p_c + \mu \frac{\partial}{\partial r} \left(r \frac{\partial v_x}{\partial r} \right)$
 SINCE $v_x(r) \rightarrow p_c = \text{CONSTANT}$ $\Rightarrow \frac{dp}{dx} = \text{CONSTANT} = \frac{p_o - p_c}{(1 - \beta)L}$

(2)

$$1 - \beta < \frac{x}{L} < 1 \rightarrow v_x = - \frac{dp}{dx} \frac{D^2}{16\mu} \left[\left(1 - \left(\frac{2r}{D} \right)^2 \right) - \frac{1 - \alpha^2}{\ln \alpha} \ln \left(\frac{2r}{D} \right) \right] \Rightarrow Q = \int_{\frac{\alpha D}{2}}^{\frac{D}{2}} 2\pi r v_x dx = \frac{p_o - p_a}{\beta L} \frac{\pi D^4}{128\mu} f(\alpha)$$

$\downarrow$
 $\frac{dp}{dx} = \frac{p_o - p_a}{\beta L}$

$f(\alpha) = 1 - \alpha^2 \left(\frac{1}{2} - \frac{1 - \alpha^2}{\ln \alpha} \right) - \frac{(1 - \alpha^2)}{\ln \alpha} \left[1 - \alpha^2 - 2\alpha^2 \ln \alpha \right]$

EQUATING (1) AND (2) GIVES

$$Q = \frac{\pi D^4}{128\mu L} \frac{p_c - p_a}{1 + \frac{\beta f}{1 - f}}, \quad p_o = p_c - \frac{(1 - \beta)(p_c - p_a)}{1 + \beta f / (1 - f)}$$